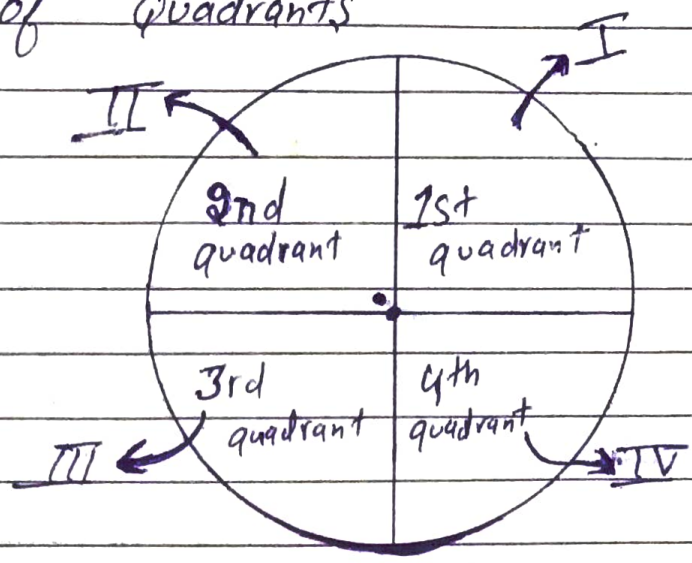
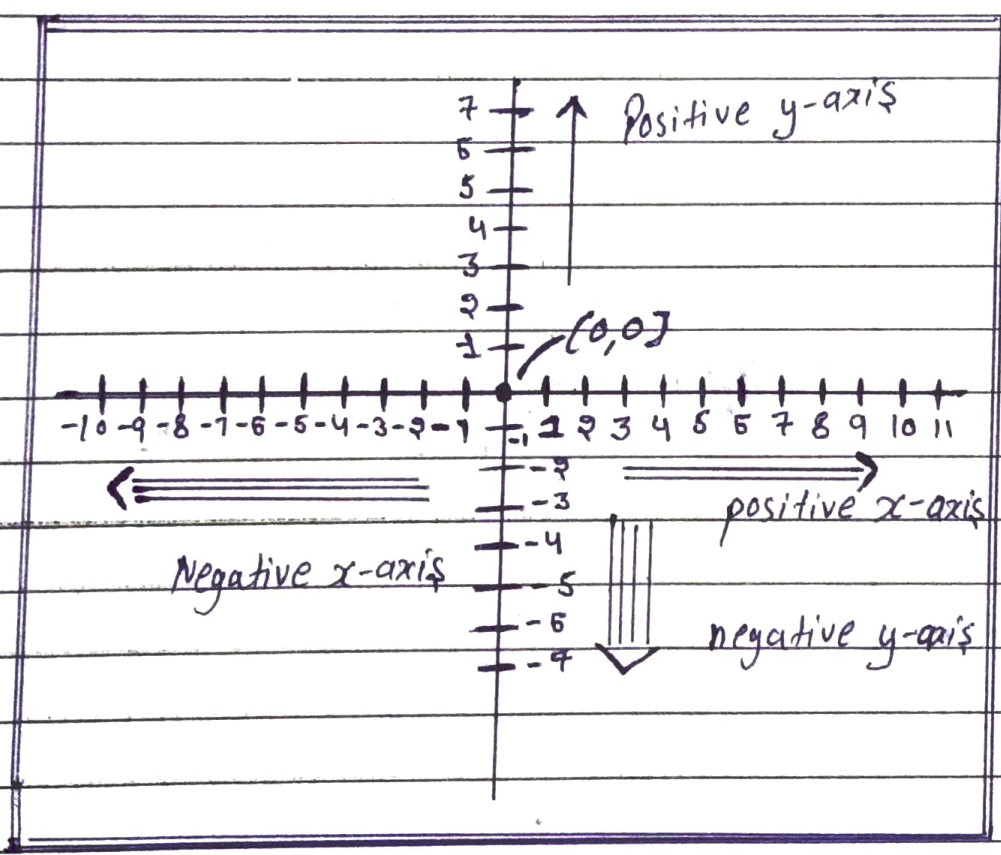


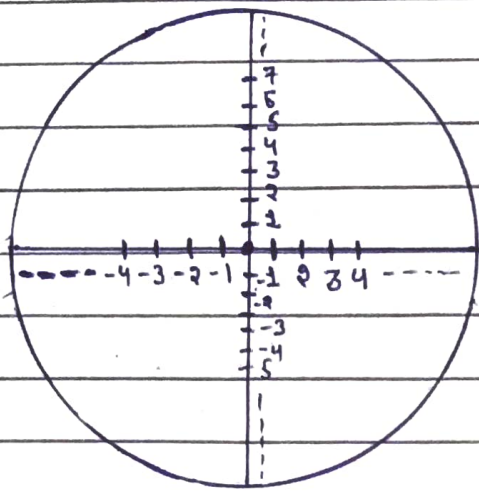
The System of Quadrants



Now, Remember the Cartesian plane, [Co-ordinate Graphs]



Keep in mind ÷ The values of trigonometric functions, values are also depends on x-y axis
 ↳ $[\sin \theta, \cos \theta, \tan \theta, \operatorname{cosec} \theta, \sec \theta, \cot \theta]$



Consider the Centre of the Circle are at Co-ordinate $(0,0)$,

→ (Remember → x and y ; both axis makes an angle of 90° (Right Angle triangle)
→ Perpendicular to each other.

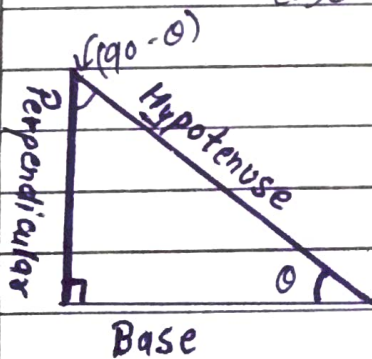
Now, for the Ist quadrant,

→ The values of Trigno-metric function will be,

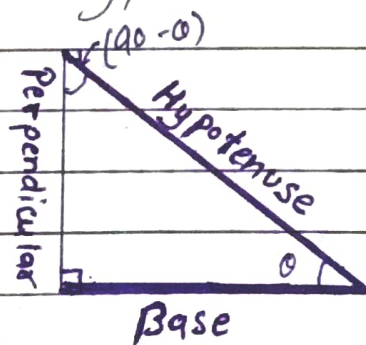
$$\sin \theta = \frac{P \text{ (Perpendicular)}}{H \text{ (Hypotenuse)}} \quad (\text{Positive})$$

$$\cos \theta = \frac{B \text{ (Base)}}{H \text{ (Hypotenuse)}} \quad (\text{Positive})$$

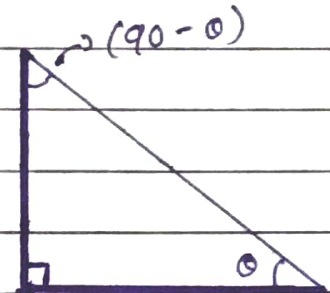
$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\text{Perpendicular (P)}}{\text{Hypotenuse (H)}} = \frac{(\text{Positive})}{(\text{Positive})} = (\text{Positive})$$



$$\sin \theta = \frac{P}{H}$$



$$\cos \theta = \frac{B}{H}$$



$$\tan \theta = \frac{P}{B} \text{ or } \frac{P}{H \times \frac{B}{H}}$$

$\tan \theta = \frac{P \times H}{B \times H}$ (So why Hypotenuse is common in Both $\sin \theta$ and $\cos \theta$, so

$$\left[\tan \theta = \frac{\sin \theta}{\cos \theta} \right]$$

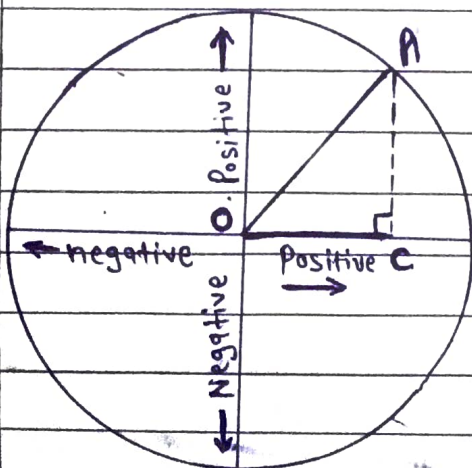
Why we are here talking about triangles?

Cause of values of Trigno-metric functions are defined by them,

Now, always remind that, in a circle,

Hypotenuse = Circle's radius

So, Cause in a circle it's radius is remains constant, so that Hypotenuse is also constant.



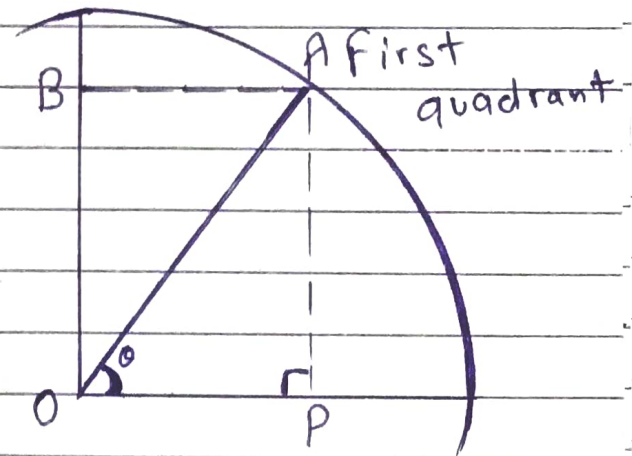
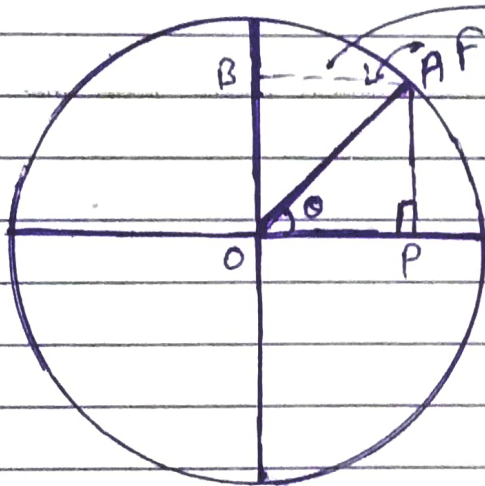
Here $OA = \text{radius}$
and, $\text{Radius} = \text{Hypotenuse}$
of $\triangle AOC$.

The values of Trigno-metric ratio are defined by the triangles.

(This would help us in understanding the Positive-Negative values [or ratios])

So,

c) The First Quadrant [F.Q]



$$AP = BO$$

And BO is along positive Y-axis.

→ Now, for $\sin \theta$,

$$\sin \theta = \frac{AP}{OA} \quad (\rightarrow \text{Perpendicular})$$

$$OA \quad (\rightarrow \text{Hypotenuse})$$

Then $\rightarrow$

AP is positive, because AP is along positive Y-axis.

OA is positive, but why ???

Remember, The Hypotenuse in a triangle can't be negative.

But why Hypotenuse $\neq$ Negative ?

Cause, Pythagoras theorem

$$\rightarrow (\text{Hypotenuse})^2 = (\text{Perpendicular})^2 + (\text{Base})^2$$

Consider Perpendicular = y
Base = x

$$x^2 + y^2 = z^2 \quad (z = \text{Hypotenuse})$$

Now, x^2

put $x = 1$ and -1
 2 and -2

$$(1)^2 = 1, (-1)^2 = 1, (2)^2 = 4, (-2)^2 = 4$$

Now, y^2

same as x^2 ,
put $y = 3, -3$
 $4, -4$

$$(3)^2 = 9, (-3)^2 = 9, (4)^2 = 16, (-4)^2 = 16$$

Outputs $\rightarrow$ for $x = [1, 1, 4, 4] \rightarrow$ Positive
for $y = [9, 9, 16, 16] \rightarrow$ Positive

even for a circular equation,

$$r = \sqrt{(x-h)^2 + (y-k)^2}$$

And the radius of a circle can't be negative,
So why that exists.

[Radius = Hypotenuse]

Now, when both OA and AP are positive
their ratio will also be positive.

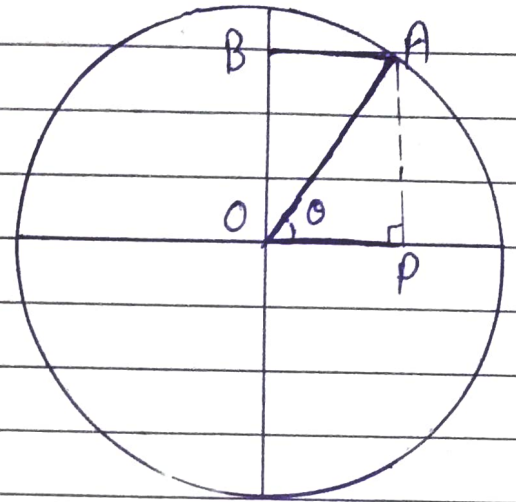
$$\text{So, } \sin \theta = \frac{AP \text{ (Positive)}}{OA \text{ (Positive)}} = \text{Positive}$$

We conclude that the $\sin \theta$ is positive in
1st Quadrant.

→ For $\cos \theta$

$$\cos \theta = \frac{OP \text{ (Base)}}{OA \text{ (Hypotenuse)}}$$

OP is along positive x-axis,
OA is Hypotenuse (Radius)
↳ Positive



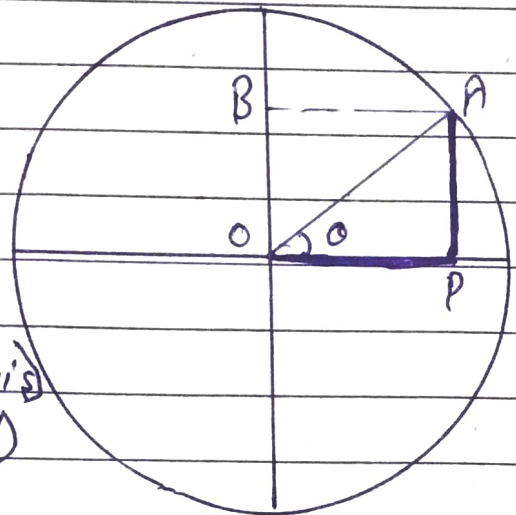
$$\cos \theta = \frac{OP \text{ (Positive)}}{OA \text{ (Positive)}} = \text{Positive}$$

$\cos \theta$ is also positive in 1st quadrant

→ For $\tan \theta$

$$\tan \theta = \frac{AP \text{ (Perpendicular)}}{OP \text{ (Base)}}$$

AP = Positive (along +ve y-axis)
OP = Positive (along +ve x-axis)



$$\tan \theta = \frac{AP \text{ (Positive)}}{OP \text{ (Positive)}} = \text{Positive}$$

$\tan \theta$ is also positive in 1st quadrant.

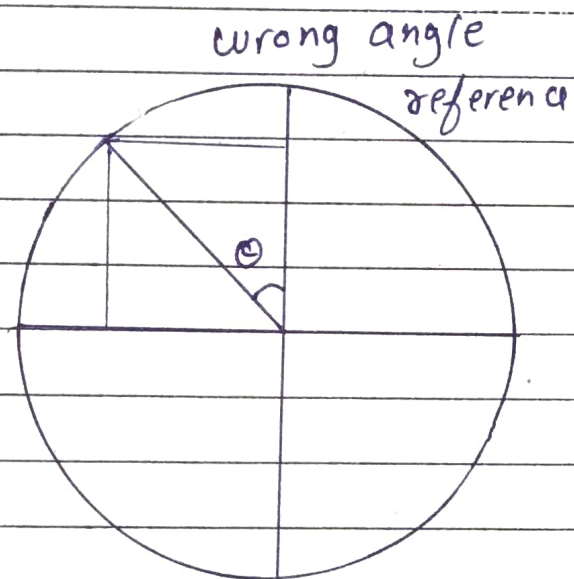
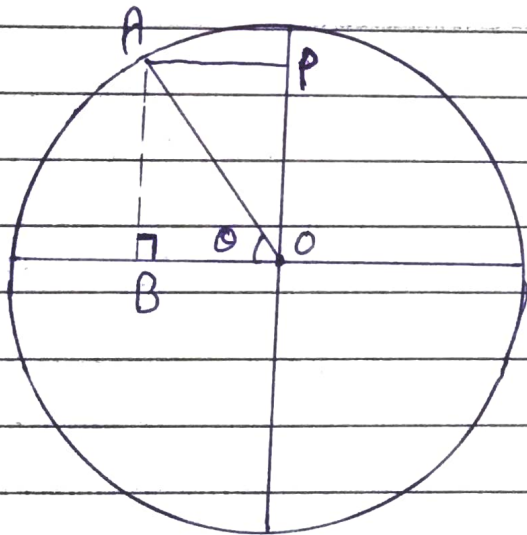
Alternatively,

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$\sin \theta$ in 1st quadrant is positive (Proved Previously)
 $\cos \theta$ in 1st quadrant is positive (Proved Previously)

$$\tan \theta = \frac{\sin \theta (\text{Positive})}{\cos \theta (\text{Positive})} = \text{Positive.}$$

Now, In 2nd Quadrant



The angle θ should be made on the x-axis both (Positive and Negative)

$$\sin \theta = \frac{AB}{OA}$$

(AB is Positive (along y-axis))
 (OA is Positive (Hypotenuse))
 (always Positive)

$$\rightarrow \sin \theta = \frac{AB \text{ (Positive)}}{OA \text{ (Positive)}} = +ve \text{ (Positive)}$$

$$\rightarrow \cos \theta = \frac{OB}{OA} \quad \begin{array}{l} \text{(Negative (along Negative x-axis))} \\ \text{(Hypotenuse (always Positive))} \end{array}$$

$$\cos \theta = \frac{OB \text{ (-ve)}}{OA \text{ (+ve)}} = -ve$$

$\cos \theta$ in Second (2nd) Quadrant is always negative

$$\rightarrow \tan \theta = \frac{AB}{OB} \quad \begin{array}{l} \text{(AB is +ve (along +ve y-axis))} \\ \text{(OB is -ve (along -ve x-axis))} \end{array}$$

$$\tan \theta = \frac{AB \text{ (+ve)}}{OB \text{ (-ve)}} = -ve$$

$\tan \theta$ in Second Quadrant is always negative

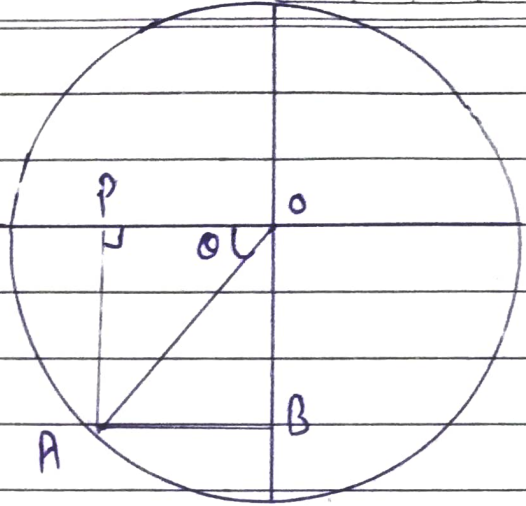
Alternating $\rightarrow \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{+ve}{-ve} = -ve$
 (cos θ is -ve in 2nd quadrant)

$\sin \theta = +ve$ $\cos \theta = -ve$ $\tan \theta = -ve$	{	$\csc \theta = +ve$ $\sec \theta = -ve$ $\cot \theta = -ve$	$\left. \begin{array}{l} \text{reciprocal of } \sin \theta \\ \text{reciprocal of } \cos \theta \\ \text{reciprocal of } \tan \theta \end{array} \right\}$
---	---	---	--

In 3rd Quadrant

AP is -ve (along -ve Y-axis)

OP is -ve (along -ve X-axis)



$$\rightarrow \sin \theta = \frac{AP}{OA} = \frac{-ve}{+ve} = -ve$$

OA (+ve)
↳ Hypotenuse

$$\rightarrow \cos \theta = \frac{OP}{OA} = \frac{-ve}{+ve} = -ve$$

OA (+ve)

$$\rightarrow \tan \theta = \frac{AP}{OP} = \frac{-ve}{-ve} = +ve$$

OP (-ve)

Since $\tan \theta$ is positive we can say that $\sin \theta$ and $\cos \theta$ must be negative, so why $\tan \theta$ is +ve.

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-ve}{-ve} = +ve$$

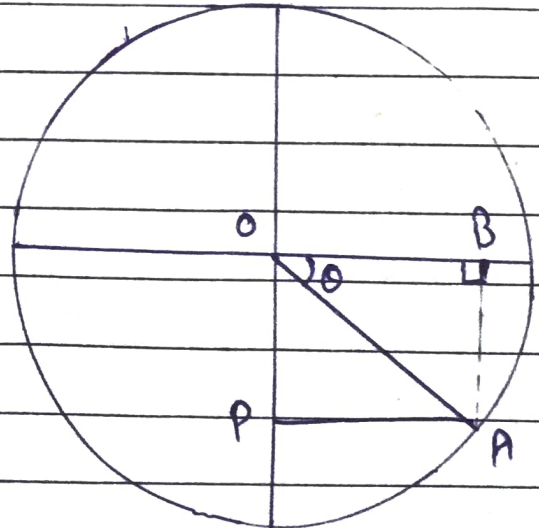
$\cos \theta$ (-ve)

In 4th Quadrant

AB = (-ve) (along -ve Y-axis)

OB = (+ve) (along X-axis) (+ve)

OA = (Hypotenuse) always (+ve)



$$\rightarrow \sin \theta = \frac{AB (-ve)}{OA (+ve)} = -ve$$

$$\rightarrow \cos \theta = \frac{OB (+ve)}{OA (+ve)} = +ve$$

$$\rightarrow \tan \theta = \frac{AB (-ve)}{OB (+ve)} = -ve$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = -ve$$

Summary $\rightarrow$

1st Quadrant: $\sin \theta \rightarrow +ve$, $\cos \theta \rightarrow +ve$, $\tan \theta \rightarrow +ve$
So why reciprocals are also +ve

2nd Quadrant: $\sin \theta \rightarrow +ve$, $\cos \theta \rightarrow -ve$, $\tan \theta \rightarrow -ve$
 $\csc \theta \rightarrow +ve$, $\sec \theta \rightarrow -ve$, $\cot \theta \rightarrow -ve$

3rd Quadrant: $\sin \theta \rightarrow -ve$, $\cos \theta \rightarrow -ve$, $\tan \theta \rightarrow +ve$
 $\csc \theta \rightarrow -ve$, $\sec \theta \rightarrow -ve$, $\cot \theta \rightarrow +ve$

4th Quadrant: $\sin \theta \rightarrow -ve$, $\cos \theta \rightarrow +ve$, $\tan \theta \rightarrow -ve$
 $\csc \theta \rightarrow -ve$, $\sec \theta \rightarrow +ve$, $\cot \theta \rightarrow -ve$

So why, Add Sugar to Coffee

$\downarrow$ All (+ve)	$\downarrow$ $\sin \theta$ (+ve)	$\downarrow$ $\tan \theta$ (+ve)	$\downarrow$ $\cos \theta$ (+ve)
---------------------------	-------------------------------------	-------------------------------------	-------------------------------------

For change one function to another
 $\sin \theta \rightarrow \cos \theta$, $\cos \theta \rightarrow \sin \theta$, $\tan \theta \rightarrow \sec \theta$
 $\sin \rightarrow \sin$, $\cos \rightarrow \cos$, $\tan \rightarrow \tan$ etc

why? do we need to convert?

→ To simplify and to derive and to know what value of function is similar to what value of the same function at a specific angle. (Numerical value)

What?

$$\text{like } \sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$$

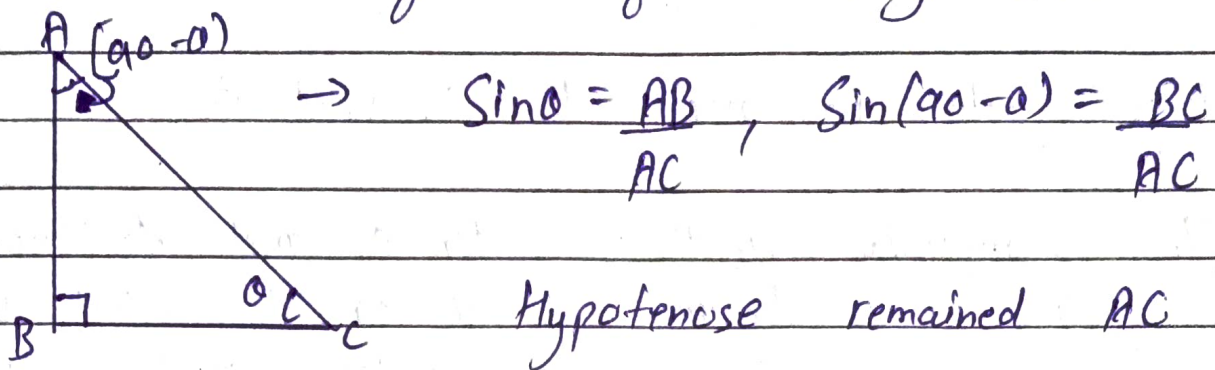
$$\text{like } \cos(150^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$$

So, $150^\circ = 30^\circ$, No brother
 $150^\circ \neq 30^\circ$

$$\text{But } \sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$$

So to know if any angle's value is similar to what simple value of function.
 And to simplify, to make it easy.

Remember one main thing, in 90° (Right angled triangle) we consider the Perpendicular, Base, Hypotenuse with the reference of ~~that~~ angle ($\angle$)



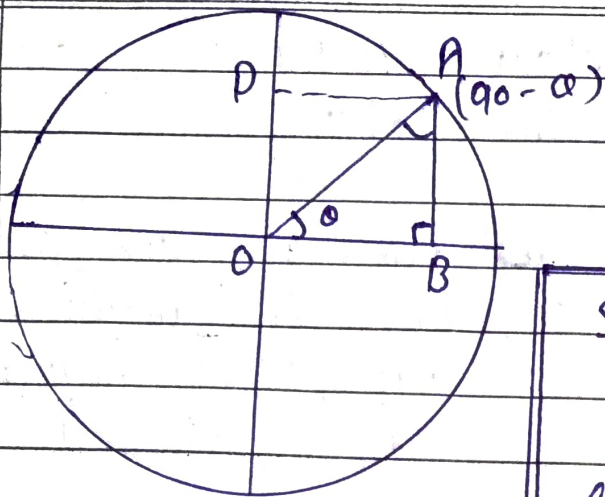
$$\cos(\alpha) = \frac{BC}{AC}, \cos(90 - \alpha) = \frac{AB}{AC}$$

By this we can say,

$$\sin(90 - \alpha) = \cos \alpha \quad \left[\because \sin(90 - \alpha) = \frac{BC}{AC} = \cos \alpha \right]$$

$$\cos(90 - \alpha) = \sin \alpha \quad \left[\because \cos(90 - \alpha) = \frac{AB}{AC} = \sin \alpha \right]$$

In a triangle $\sin \alpha$ and $\cos \alpha$ are related to each other.



always keep in mind about the positive-negative axis of $x-y$.

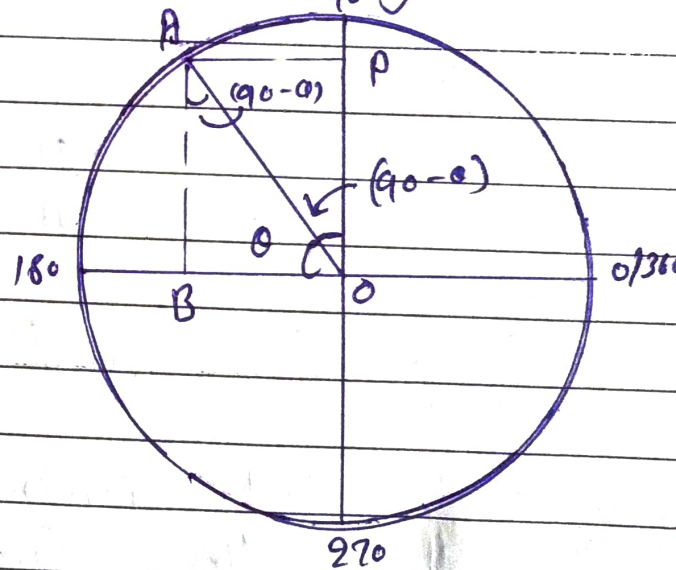
$$\sin \theta = \frac{AB}{OA} = \cos(90 - \theta)$$

$$\cos \theta = \frac{OB}{OA} = \sin(90 - \theta)$$

always keep in mind about the reference of angle by which P and B and H are selecting.

But in 2nd Quadrant $\rightarrow$

$\rightarrow$ But 2nd Quadrant mean on a full 180° rotation



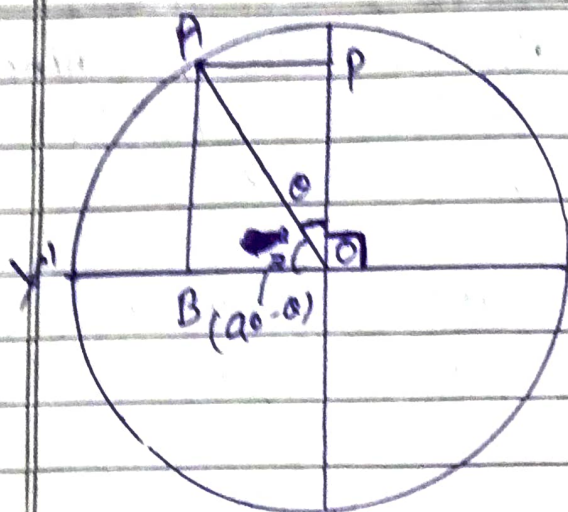
$$\sin(180 - \theta) = \frac{AB}{OA}$$

$$\cos(90 - \theta) = \frac{OP}{OA} = \frac{AB}{OA} = \sin(180 - \theta)$$

Now, $\sin(180 - \theta) = \sin \theta$

$$\sin(180 - \theta) = \sin(\theta)$$

($\sin \theta$ is positive in 2nd quadrant)



$$\angle X'OA = 90 + \theta$$

$$\sin(90 + \theta) = ?$$

x' / ~~remember~~ remember the reference of θ , but Sin will remain AB, that will not be OA changed

But to make it simple we gonna take θ in place of $\sin(90 - \theta)$
 So, we have to see which is similar to
 $\sin(90 - \theta) = \frac{AB}{OA}$

$$\cos(\theta) = \frac{OP}{OA} = \frac{AB}{OA} \quad (OP = AB)$$

$$\text{So, } \sin(90 + \theta) = \cos(\theta)$$

just remember the value of actual sin
 Here $\sin(90 - \theta) = \frac{AB}{OA}$

But we have to find $\sin(90 + \theta)$, so we find $\cos \theta$ which can give value of $\sin(90 - \theta)$, cause $\sin(90 - \theta)$ will be the actual value of Sin.
 That's why.

You can apply these two methods in other Quadrants for Sin and Cos and $\tan = \frac{\sin}{\cos}$ (derive it)